



Idiopathic hypersomnia is a 24-hour disorder

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Received: 24 October 2025 / Revised: 24 October 2025 / Accepted: 4 November 2025
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Abstract

Idiopathic hypersomnia is a heterogeneous disorder with daytime and nighttime symptoms that have a profound 24-hour effect on individuals' well-being. Daytime signs and symptoms include excessive daytime sleepiness, unrefreshing naps, autonomic dysfunction, fatigue, and brain fog; nighttime signs and symptoms include long sleep, and sleep inertia. Symptom overlap between idiopathic hypersomnia and narcolepsy type 2 and the varied disease course of idiopathic hypersomnia present challenges in diagnosis. Furthermore, the diagnostic criteria for idiopathic hypersomnia are inadequate in their current form, and their use can lead to misdiagnosis and suboptimal treatment. The unclear pathophysiology of idiopathic hypersomnia and the absence of reliable biomarkers necessitate careful use of a variety of tests to make an accurate diagnosis. The Idiopathic Hypersomnia Severity Scale captures symptoms that distinguish idiopathic hypersomnia from narcolepsy, but this scale requires further evaluation in real-world populations and should be used with objective measures of sleep parameters and other validated instruments to accurately capture the full range of symptoms over a 24-hour period. This review describes a holistic view of the symptoms and impact of idiopathic hypersomnia, and frames diagnostic challenges and treatment considerations within the context of this view. Expert clinical opinion that aims to improve the diagnostic accuracy and treatment of idiopathic hypersomnia is provided.

Keywords Idiopathic hypersomnia · Sleepiness · Diagnosis · Symptoms

Introduction

Idiopathic hypersomnia is a rare and debilitating central hypersomnolence disorder with an estimated US prevalence up to 0.01% [1–3]. Symptoms have their onset during adolescence and early adulthood and affect females more than males [4, 5]. Identification and definitive diagnosis of idiopathic hypersomnia are challenging because its symptoms and disease course are heterogeneous and the symptoms overlap those of other central hypersomnolence disorders

[1, 2]. The pathophysiology of idiopathic hypersomnia is unclear, and biomarkers for definitive diagnosis have yet to be identified. In addition, approaches to diagnosis and assessment of individuals with idiopathic hypersomnia vary across institutions and countries.

Excessive daytime sleepiness (EDS) is a key symptom of idiopathic hypersomnia but is also a primary characteristic of narcolepsy type 1 (NT1) and type 2 (NT2) and is observed in several other conditions [1, 6]. Other characteristics that may overlap between idiopathic hypersomnia and NT2 include Multiple Sleep Latency Test (MSLT) mean sleep latency ≤ 8 minutes, presence of sleep inertia, and absence of cataplexy [1]. A finding of < 2 sleep-onset rapid eye movement periods (SOREMPs) within 15 minutes of sleep onset on polysomnography (PSG) prior to the MSLT, and/or during the MSLT, differentiates idiopathic hypersomnia from NT2 [1, 2]. In addition to having EDS, individuals with idiopathic hypersomnia may experience an excessive need to sleep, which may result in nighttime long sleep time (LST) ≥ 10 hours, reported in 50% or more of individuals [7–9]. Other common symptoms are sleep inertia, unrefreshing naps (> 1 hour), fatigue, brain fog, and

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autonomic dysfunction [10–14]. Individuals with idiopathic hypersomnia therefore experience symptoms continually, night and day, which can have a profoundly negative impact on physical and mental well-being, daily activity, work productivity, and social relationships [15–17].

Some of the tools that are regularly used to diagnose idiopathic hypersomnia and measure treatment effectiveness either are unreliable or provide an incomplete picture of symptoms. MSLTs repeated over time have highly variable results [18–20]. Over several decades, the Wisconsin Sleep Cohort study evaluated cases of probable idiopathic hypersomnia based on MSLT, PSG, and Epworth Sleepiness Scale (ESS) score results in the general population, with MSLTs and PSG administered approximately every 4 years [21]. According to longitudinal results, 50% of participants with probable idiopathic hypersomnia demonstrated increased MSLT mean sleep latency (> 8 minutes) between their first and second evaluations, performed an average of 3.6 years apart [21]. Despite having increased MSLT mean sleep latency, most of these participants continued to register ESS scores ≥ 11 [21]. Symptom remission, defined as an ESS score < 11, was observed in 40% of participants [21]. The ESS is widely used in clinical settings, as EDS occurs in 5%–33% of individuals in the general population and across several different medical conditions [6, 22–24]. However, the utility of the ESS among patients with idiopathic hypersomnia is limited, as it does not assess symptoms of sleep inertia, LST, and need for daytime naps [25].

The Idiopathic Hypersomnia Severity Scale (IHSS) was developed to assess EDS and other daytime and nighttime symptoms in idiopathic hypersomnia [26]. The 14-item IHSS captures symptoms across three domains: daytime functioning (7 items), long sleep duration and sleep inertia (5 items), and napping (2 items) [9]. The intensity, frequency, and impact of symptoms are rated on a Likert scale ranging from 0–3 or 0–4, providing a total score ranging from 0–50 [26]. IHSS total score ranges of 0–12, 13–25, 26–38, and 39–50 are used to classify mild, moderate, severe, and very severe disease, respectively. [9] A score of 22 was deemed the optimal threshold for differentiating untreated individuals with idiopathic hypersomnia from healthy individuals [26]. Comparisons of IHSS total scores in individuals with idiopathic hypersomnia who were treatment-naïve and those who were treated determined a minimum clinically important difference (MCID) of 4 points [9].

Idiopathic hypersomnia is characterized by symptoms that can occur throughout the 24-hour period and substantially impact individuals' lives (Fig. 1). However, awareness and appreciation of the 24-hour nature of idiopathic hypersomnia may be incomplete in the clinical community. Thus, this review elucidates the 24-hour nature of idiopathic hypersomnia symptoms and the impact of these

symptoms on health-related quality of life (HR-QoL). Awareness of this around-the-clock disorder may help improve diagnostic accuracy, inform appropriate treatment choices, and provide a better understanding of the effects of treatment on symptom management. Given the complexity and heterogeneity of idiopathic hypersomnia, accurate diagnosis is essential in ensuring that patients receive timely, appropriate treatment for their daytime and nighttime experience.

Methods

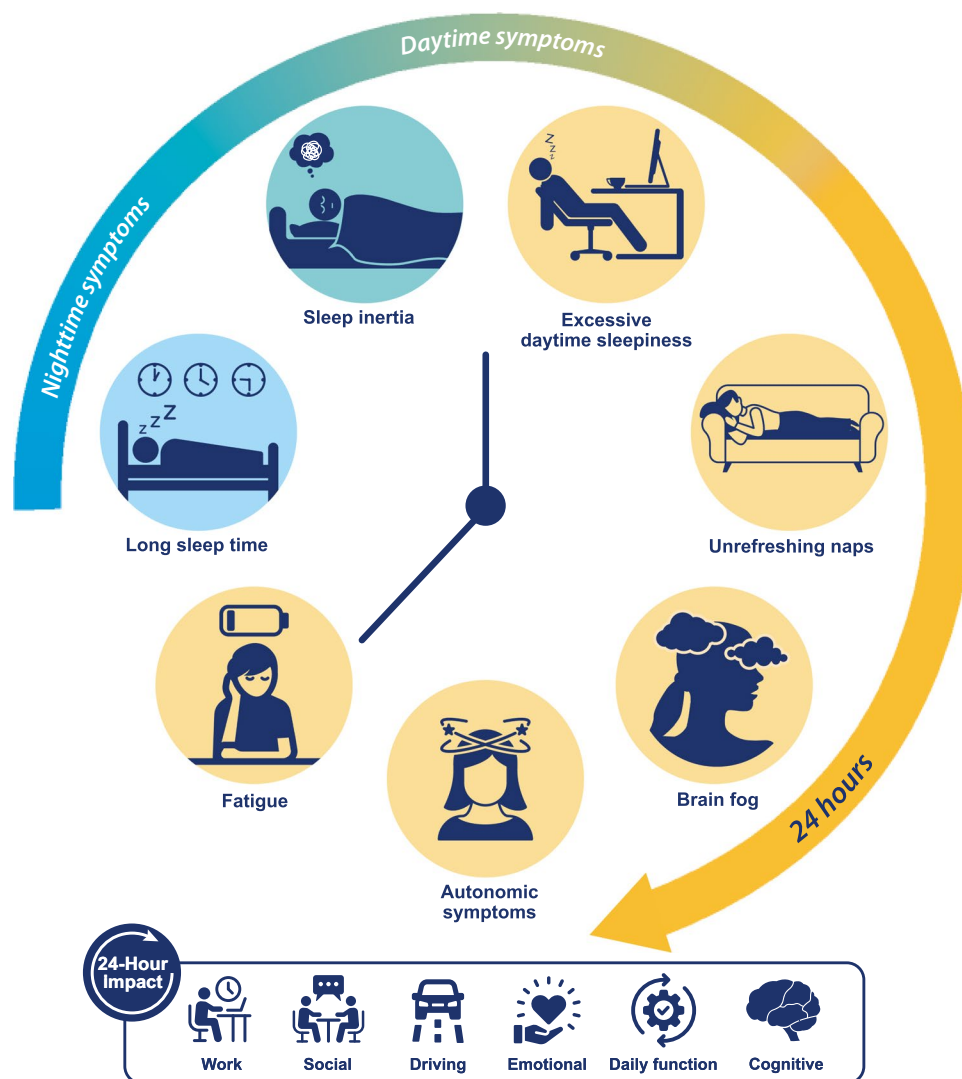
To develop this narrative review, PubMed was searched for “idiopathic hypersomnia” and a variety of other terms associated with symptoms of idiopathic hypersomnia, with no publication date limitation. Reference lists of identified articles were mined for additional relevant publications. Clinical and real-world study articles, review articles, and expert opinion articles were reviewed.

Daytime signs and symptoms

Excessive daytime sleepiness

The *International Classification of Sleep Disorders – Third Edition, Text Revision* (ICSD- 3-TR) includes EDS as a requirement for idiopathic hypersomnia diagnosis [2]. Although the vast majority of individuals with idiopathic hypersomnia experience EDS, a minority report excessive need to sleep rather than EDS, based on expert opinion. It is unclear if this group truly represents idiopathic hypersomnia or long sleepers. In the Hypersomnia Foundation Registry (HFR) study, 98% of individuals with daily sleep duration ≥ 10 hours and 97% of individuals with daily sleep duration < 10 hours reported EDS; among those treated with off-label medications, 64% still reported EDS [8]. In the US-based Real-World Idiopathic Hypersomnia Outcomes Study (ARISE), 89% of individuals with daily sleep duration ≥ 11 hours and 87% of individuals with daily sleep duration < 11 hours had ESS scores > 10, indicating pathological levels of sleepiness, despite most participants (89%) being treated with off-label medications [27]. Evaluation of circadian behavior patterns in a large cohort study demonstrated that, compared with healthy controls, patients with idiopathic hypersomnia were more alert in the evening versus the morning; feeling most tired in the morning was indicated by 85% of patients with idiopathic hypersomnia and 48% of healthy controls [28]. IHSS items 6 and 9 are used to capture EDS [9].

Fig. 1 Daytime and nighttime symptoms in idiopathic hypersomnia and their impact on quality of life [7, 12, 13, 15]



Naps

Naps are common among individuals with idiopathic hypersomnia and are captured by IHSS items 6 and 7 [9]. Across studies, mean daily nap duration ranges from <0.5 to 2.8 hours [4, 8, 14, 28]. Findings from the HFR study showed that individuals with idiopathic hypersomnia and LST (nighttime sleep duration ≥ 10 hours) were more likely to intentionally take daytime naps (64.2%) and had longer mean nap duration (2.8 hours) than those without LST (42.1% and 2.0 hours) [8]. However, unrefreshing naps are frequently reported by individuals with idiopathic hypersomnia [4, 7, 14, 28, 29], particularly those with long sleep duration [7], whereas individuals with NT2 typically report daytime naps that are brief (<30 minutes) and restorative [1, 30–32]. In one study, however, based on the IHSS, only 37% of patients with NT2 endorsed feeling refreshed after napping [33]. In addition, daytime sleepiness in idiopathic hypersomnia can vary significantly, with some people

experiencing short, refreshing naps and others experiencing longer, nonrestorative naps with a continuous feeling of drowsiness [34].

Brain fog and cognitive impairment

Brain fog encompasses multiple self-reported symptoms, including difficulty concentrating, forgetfulness, grogginess, cognitive difficulty and slowness, difficulty communicating, and fatigue [13, 35]. In idiopathic hypersomnia, brain fog is reported as the inability to think clearly or concentrate throughout the day [8, 13]. A prospective, observational cohort study found several symptoms present in a substantially higher percentage of individuals with idiopathic hypersomnia versus healthy controls, respectively: memory problems (79% vs 43%), frequent forgetfulness (47% vs 5%), attention deficit (55% vs 18%), and mind going blank (58% vs 13%) [28]. The duration of maximum concentration was significantly shorter among patients with idiopathic

hypersomnia versus healthy controls (mean, 1 vs 3.4 hours) [28]. Findings from the HFR study demonstrated that >80% of participants with idiopathic hypersomnia experienced brain fog [8]. In the ARISE study, brain fog was reported to be one of the idiopathic hypersomnia symptoms that are most difficult to treat [27]. In hypersomnolence disorders, it is unclear whether brain fog indicates an alertness deficit caused by sleepiness, or is a distinct symptom that is independent of sleepiness [13]. Furthermore, the experience of brain fog may be akin to that of other cognitive symptoms, such as mental fatigue and cognitive dysfunction [13]. The brain fog descriptions offered by individuals with idiopathic hypersomnia in the HFR study likely reflect slow or sluggish thinking that is variable, occurring anytime and irrespective of exertion [13]. A recent meta-analysis of studies evaluating cognitive impairment in hypersomnia disorders found that, compared with matched controls, individuals with idiopathic hypersomnia had substantial impairment in sustained attention [36].

Autonomic dysfunction

Autonomic dysfunction in individuals with idiopathic hypersomnia comprises orthostatic, vasomotor, gastrointestinal, bladder, secretomotor, and pupillomotor symptoms [12]. Symptoms of autonomic dysfunction may manifest as temperature dysregulation, Raynaud syndrome or cold extremities, dizziness, feeling faint, heart palpitations, digestive problems (eg, diarrhea, constipation), eye or vision problems, allergy, and frequent headaches [12, 28, 37]. While the etiology of autonomic dysfunction associated with idiopathic hypersomnia is unclear, a possible explanation is that excessive sleepiness and fatigue reduce physical activity and conditioning, which may predispose these individuals to autonomic dysfunction and orthostatic intolerance [12, 38]. An alternative explanation is that there is a common pathophysiological mechanism, with autonomic dysfunction inherent to idiopathic hypersomnia [12]. Findings from an online cohort study of individuals with idiopathic hypersomnia showed that autonomic symptom burden was greatest in the orthostatic and vasomotor domains [12]. The same study found that the most common comorbidity in individuals with idiopathic hypersomnia was postural orthostatic tachycardia syndrome, which mainly affects young or middle-aged women with similar autonomic symptoms [12, 38]. The prevalence of autonomic dysfunction is not established in adults, although in a study of pediatric patients, autonomic dysfunction was present in 11/18 (61%) of those with idiopathic hypersomnia, 11/46 (24%) with NT1, and 7/17 (41%) with NT2 [39].

Fatigue

Fatigue is common among individuals with idiopathic hypersomnia [17]. A single-center study found comparable levels of fatigue, assessed with the Fatigue Severity Scale (FSS), among patients with idiopathic hypersomnia, NT1, and NT2 [11]. Among patients with idiopathic hypersomnia, FSS scores were significantly higher (indicating worse fatigue) for those with sleep duration ≥ 11 hours than for those with sleep duration <11 hours. [11] In another single-center study, fatigue was found to be a significant determinant of HR-QoL in patients with idiopathic hypersomnia [17].

Nighttime signs and symptoms

Long sleep time

While the ICSD-3-TR diagnostic criteria do not differentiate between LST and non-LST subgroups, a 24-hour sleep duration of 11 hours or more (from 24-hour PSG or 7-day wrist actigraphy with a sleep log) may count toward diagnosis in lieu of a mean sleep latency ≤ 8 minutes on the MSLT [2]. It is important to note that individuals who have mean sleep latency ≤ 8 minutes on the MSLT may also be long sleepers, as an inverse correlation was demonstrated between sleep latency on the second MSLT and sleep duration on PSG using a 32-hour bed rest protocol [40]. A cutoff of ≥ 10 hours for long nighttime sleep in individuals with idiopathic hypersomnia is often used in the literature, based on a general definition of “long sleepers” [34]. Approximately half the individuals with idiopathic hypersomnia in the HFR and ARISE studies had nighttime sleep duration ≥ 10 hours [8, 27]. In several US studies, approximately 50% of individuals with idiopathic hypersomnia are reported to have LST [7, 8, 27]. In other studies of idiopathic hypersomnia, the proportion of participants with nighttime sleep duration ≥ 10 hours is more varied (range, 19%–100%) [4, 26, 29, 41–43]. In a French study evaluating the responsiveness of the IHSS to medications in patients with idiopathic hypersomnia, 62%–75% of treated patients and 77%–82% of untreated patients reported their ideal nighttime sleep duration was ≥ 9 hours, as capture by IHSS item 1 [9]. However, some patients (such as those with nonspecified hypersomnia, or those with idiopathic hypersomnia without LST) may overestimate the amount of time they spend sleeping, while other patients (those with idiopathic hypersomnia with LST) may underestimate this duration [44]. Therefore, objective assessment of sleep duration is critical, and can be accurately determined with extended PSGs. French consensus recommendations

emphasize the importance of a PSG test of ≥ 24 hours in people with sleep latencies > 8 minutes on the MSLT [45].

Certain patient clinical and demographic characteristics have been associated with LST. Higher IHSS total scores and greater symptom severity have been associated more with nighttime sleep duration > 9 hours than with nighttime sleep duration ≤ 9 hours and multivariate analyses indicated a significant association between prolonged nighttime sleep and depression [9]. A prospective single-center study demonstrated that patients who had idiopathic hypersomnia with LST (> 10 hours total sleep time) were younger than those without LST (mean ages of 29 years and 40 years, respectively) and had lower body mass index (means of 23 kg/m² and 26 kg/m², respectively) [42]. In another single-center study, patients who had idiopathic hypersomnia with LST (> 10 hours total sleep time) were more likely than their non-LST counterparts to be female; however, the non-LST subgroup had higher ESS scores at baseline [7]. In addition, higher levels of fatigue and autonomic dysfunction were found in the LST subgroup compared with the non-LST subgroup [7].

Sleep inertia

Sleep inertia is extreme difficulty waking from sleep (sometimes called “sleep drunkenness” in the literature) [46]. The effects of sleep inertia last from a few minutes to a few hours, and can be quite severe in people with idiopathic hypersomnia, accompanied by ataxia, orthostatic disequilibrium, and hyporeflexia [47]. While sleep inertia is frequently observed with idiopathic hypersomnia, it is not specific to or essential for diagnosis, as sleep inertia is also observed in 20%–55% of individuals with NT2 [47, 48]. In 12%–27% of individuals with idiopathic hypersomnia, sleep inertia is accompanied by automatic (unconscious) behavior, disorientation, confusion, irritability, and poor coordination [48]. As such, brain fog may follow sleep inertia in the continuum of nighttime and daytime symptoms. Sleep inertia is captured by IHSS items 3–5 [9]. In a study evaluating the responsiveness of the IHSS to idiopathic hypersomnia treatment, 88%–91% of untreated patients and 78%–86% of treated patients reported sleep inertia [9].

The Sleep Inertia Questionnaire (SIQ), recently validated in patients with central disorders of hypersomnolence, covers cognitive, behavioral, physiological, and emotional effects of sleep inertia [49]. An SIQ sum score of 42 distinguishes individuals with central disorders of hypersomnolence from controls, and a cutpoint of 25 minutes for sleep duration symptoms distinguishes idiopathic hypersomnia from NT1 [49]. In contrast, no SIQ parameters are able to distinguish idiopathic hypersomnia from NT2 or subjective EDS [49]. The visual analog scale for sleep inertia (VAS-SI)

directs individuals to mark a point on a 100-mm line indicating how difficult or how easy it is to wake up (ranging from *very easy* to *very difficult*) [41, 50]. The VAS-SI was implemented in a phase 3, placebo-controlled, double-blind, randomized, withdrawal study of low-sodium oxybate (LXB) in participants with idiopathic hypersomnia [41]. A post hoc analysis of efficacy outcomes from the phase 3 study found a high correlation between VAS-SI and IHSS item 3 (difficulty waking up) and item 4 (time taken to become fully physically and intellectually functioning) scores related to sleep inertia, and identified a minimum clinically important difference of 10–12 mm on the VAS-SI [50]. The Psychomotor Vigilance Task (PVT) is a simple reaction-oriented task where individuals are asked to quickly press a button in response to a visual stimulus [51, 52]. Lapses in PVT response times can serve as an objective measure of sleep inertia [53]. Among patients with idiopathic hypersomnia in a single-center study, results showed more lapses and a significant decrease in mean slowest response speed for those with severe sleep inertia compared with those without sleep inertia [53].

Sleep architecture alterations

Sleep architecture has not been well-characterized in people with idiopathic hypersomnia. Multiple studies have reported greater sleep efficiency in patients with idiopathic hypersomnia than in subjectively sleepy controls [54, 55], although these findings are inconclusive [56, 57]. Decreases in the percentage of time spent in slow-wave sleep have also been shown in patients with idiopathic hypersomnia compared with healthy [56] and subjectively sleepy controls [55]. However, conflicting results have been reported in a retrospective study that found no difference in the percentage of time spent in any sleep stage [58]. Sleep architecture remains an area to further explore in people with idiopathic hypersomnia.

Impact of idiopathic hypersomnia on health-related quality of life

The daytime and nighttime symptoms of idiopathic hypersomnia have a profound effect on HR-QoL. Compared with individuals without sleepiness-related impairment, a clinical reference population, healthy controls, and normative data, individuals with idiopathic hypersomnia demonstrate impairment across physical, mental, emotional, and social functioning domains [12, 15–17, 59, 60]. In a study of untreated and treated patients with idiopathic hypersomnia, an increase in IHSS total scores significantly increased the risk of moderate or severe depression (Beck Depression

Inventory II scores ≥ 20) and poor HR-QoL (EQ-5D visual analog scale scores < 60) [9].

Decreased alertness, difficulty awakening, and cognitive dysfunction reduce the ability of individuals with idiopathic hypersomnia to perform well at work or school [60]. In the ARISE study, employed individuals with idiopathic hypersomnia had a high degree of work productivity impairment (51.4%), defined as a combination of the absenteeism (work time missed) and presenteeism (impairment while working) scores on the Work Productivity and Activity Impairment Questionnaire: Specific Health Problem [15, 61]. In addition, individuals with idiopathic hypersomnia face restricted eligibility for certain occupations and risk being dismissed or forced to relocate [60]. Social relationships and activities are also impaired in individuals with idiopathic hypersomnia, who report feeling stigmatized because of their illness [15, 59, 62].

Severe EDS and lack of alertness increase the risk for crashes among drivers with idiopathic hypersomnia [59, 60, 63]. A Japanese study showed that, among patients with idiopathic hypersomnia who had a driver's license or were accustomed to driving, 55% had been in an accident or nearly missed being in an accident within a 5-year period [59]. Similarly, a French cross-sectional study showed that, regardless of symptomatic treatment, individuals with idiopathic hypersomnia or narcolepsy had a higher risk for car accidents than healthy controls [63]. However, accident risk for patients treated for > 5 years was similar to that for healthy controls [63]. Notably, the risk for car accidents was higher in individuals with idiopathic hypersomnia and NT2 than in those with NT1 [63].

Confirming a diagnosis of idiopathic hypersomnia

For adult patients with idiopathic hypersomnia, an accurate diagnosis is essential in guiding clinicians' treatment selections and improving outcomes. For pediatric patients with idiopathic hypersomnia, currently there are no diagnostic criteria, as data supporting an abnormal threshold for "long sleep time" in children are lacking. Although the ICSD-3 text for idiopathic hypersomnia was recently revised (ICSD-3-TR), no changes were made to the actual diagnostic criteria [2, 64]. In clinical practice, diagnosis continues to be based primarily on a finding of EDS, with other disorders ruled out [2, 64]. Generally, there is no systematic assessment of sleep inertia in clinical practice, although the subjective assessments IHSS, SIQ, and VAS-SI may be considered. Multiple clinical scenarios may make it difficult to confirm a diagnosis of idiopathic hypersomnia; potential avenues for diagnosis are illustrated in Fig. 2.

According to current ICSD-3-TR criteria, a diagnosis of idiopathic hypersomnia requires either mean sleep latency ≤ 8 minutes on the MSLT or total sleep time ≥ 11 hours on 24-hour PSG or actigraphy monitoring combined with a sleep log averaged over a period of ≥ 7 days with unrestricted sleep [2]. These assessments come with caveats, including variability in MSLT results over time and, in the United States, usual PSG durations of 6–7 hours (which cannot capture the diagnostic criterion of LST ≥ 11 hours) and, despite American Academy of Sleep Medicine (AASM) recommendations [65], limited use of actigraphy because of lack of reimbursement by insurance companies.

Over a period of years, an individual's MSLT findings may change as a result of variation in disease severity (with remission spontaneously occurring in some cases) [29, 46, 66, 67]. Daytime sleepiness, which may be influenced by other comorbid health conditions, may change (increase or decrease) as well [68]. In individuals with idiopathic hypersomnia or NT2, results of repeat MSLTs have been inconsistent, leading to misdiagnosis [18–20]. In addition, a substantial proportion of patients with idiopathic hypersomnia have MSLT latencies > 8 minutes [42, 69, 70]. Thus, idiopathic hypersomnia cannot be diagnosed with the MSLT and instead must meet the alternative criterion of ≥ 11 hours of nighttime sleep, as recorded by PSG or actigraphy (further discussed later in this section) [2].

An extended PSG study ensures that an accurate diagnosis is possible in individuals with normal MSLT findings. In the United States, 24-hour PSG studies often are not ordered because of the cost associated with extended technical support and laboratory utilization, which can lead to insurance billing obstacles. However, a single extended PSG that leads to an early diagnosis of idiopathic hypersomnia can be cost-saving in the long term by avoiding expenditures associated with less effective diagnostic measures (eg, magnetic resonance imaging, positron emission tomography or genetic tests) and repeat PSG-MSLT testing [71]. In an era of precision therapeutics, an accurate diagnosis that identifies LST may be necessary to avoid exposure to, and costs associated with, suboptimal medications.

In individuals with normal MSLT results, LST may be captured by actigraphy with a sleep diary, plus the required diagnostic elements of the PSG study. AASM practice guidelines recommend using actigraphy in addition to PSG to monitor total sleep time before MSLT administration [65]. This approach also ensures that individuals receive adequate pre-MSLT nighttime sleep, which improves the accuracy of MSLT findings [65]. Actigraphy device setting parameters are critical for accurate estimation of sleep; however, standardized actigraphic parameters have not been established for evaluating total sleep time in individuals with idiopathic hypersomnia [72]. A study of the Actiwatch 2 (Philips

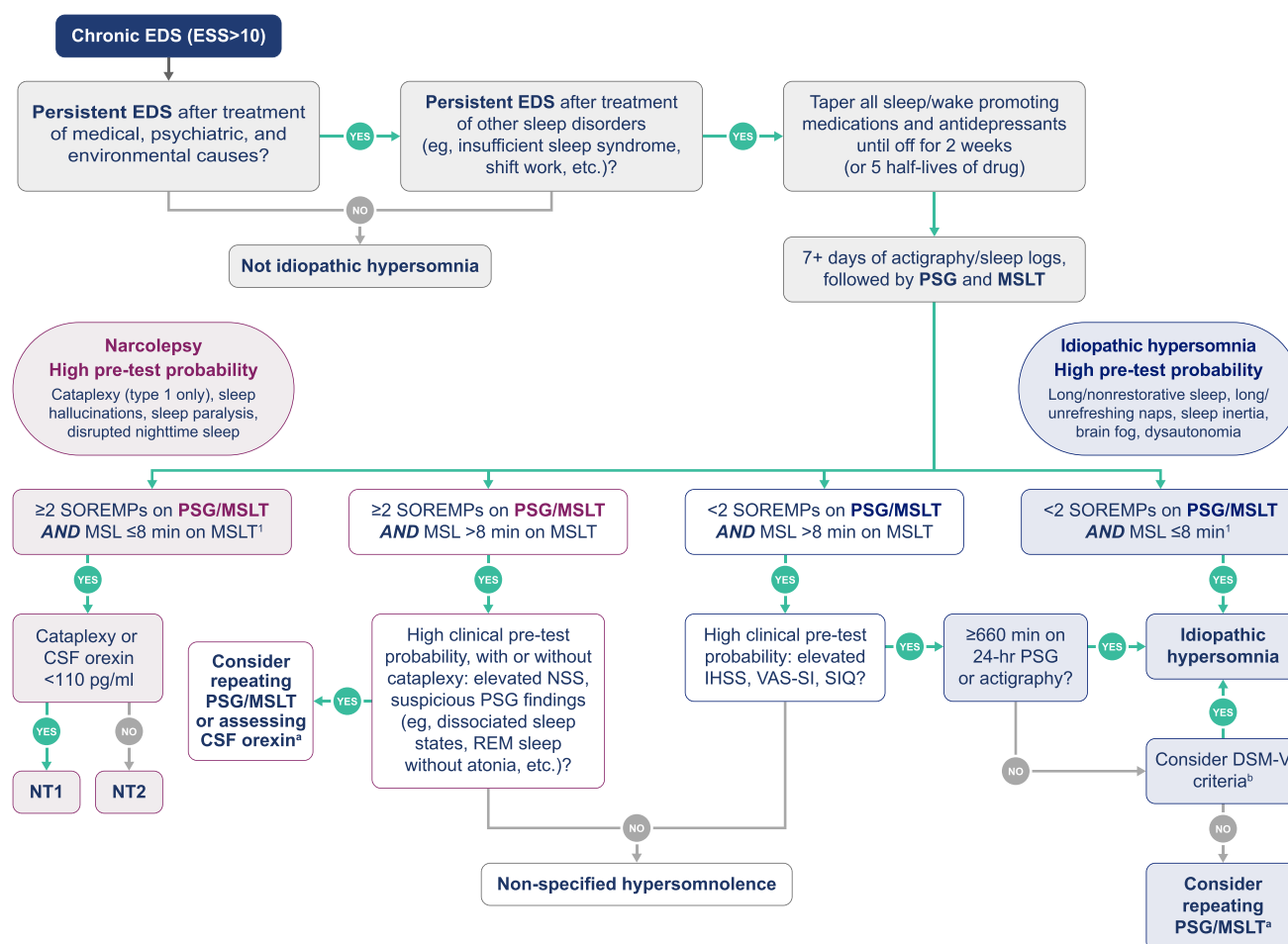


Fig. 2 Clinical and neurophysiological pathways to confirm a diagnosis of idiopathic hypersomnia or narcolepsy. ^aConsider repeating PSG/MSLT evaluations, or evaluating actigraphy, if there are any concerns regarding validity of previous tests (eg, inability to fully taper off sleep/wake promoting medications or antidepressants, insufficient sleep prior to testing) or if individual has a high pre-test probability. ^bIf there still is a strong suspicion for idiopathic hypersomnia, despite the lack of satisfying the ICSD-3-TR criteria, the more lenient DSM-V criteria (ie, unspecified hypersomnolence disorder) can be considered for potential diagnosis. *DSM-V*, *Diagnostic and Statistical Manual of*

Mental Disorders, Fifth Edition; *EDS*, excessive daytime sleepiness; *ESS*, Epworth Sleepiness Scale; *hr*, hour; *ICSD-3-TR*, *International Classification of Sleep Disorders*, Third Edition, Text Revision; *IHSS*, Idiopathic Hypersomnia Severity Scale; *min*, minute; *MSLT*, mean sleep latency; *MSLT*, multiple sleep latency test; *NSS*, Narcolepsy Severity Scale; *NT1*, narcolepsy type 1; *NT2*, narcolepsy type 2; *PSG*, polysomnography; *REM*, rapid eye movement; *SIQ*, Sleep Inertia Questionnaire; *SOREMP*, sleep onset rapid eye movement sleep period; *VAS-SI*, Visual Analog Scale for Sleep Inertia

Respironics, Murrysville, PA) device demonstrated that combining the low sleep–wake activity threshold parameter with the 25-epoch sleep-immobility onset-and-offset parameter provided the optimal estimate of total sleep time in individuals with idiopathic hypersomnia [72]. A recent study that compared results from actigraphy monitoring and PSG in drug-naïve individuals with idiopathic hypersomnia found that conventional actigraphic algorithms may overestimate these individuals' total sleep time [73]. However, a more recent study discovered a novel actigraphy algorithm using deep learning yielded fewer errors than previous algorithm studies in a large sample of patients with suspected idiopathic hypersomnia [74]. Thus, determination of LST based on an 11-hour threshold from actigraphy alone should

be interpreted with caution in individuals with idiopathic hypersomnia [73].

Symptom overlap between idiopathic hypersomnia and NT2 should be considered during initial patient evaluation. In a comprehensive overview of the symptoms of idiopathic hypersomnia compared with those of NT1 and NT2, Dauvilliers and colleagues elucidated the symptom overlap between idiopathic hypersomnia and NT2 [1]. Notably, in a retrospective single-center study, 59% of patients with untreated NT2 reported nighttime sleep ≥ 9 hours (with 31% of those reporting durations ≥ 11 hours) and 83% reported sleep inertia, based on the IHSS [33]. The Narcolepsy Severity Scale–2 (NSS-2), which captures symptoms of EDS and symptoms (disturbed nocturnal sleep,

hallucinations, sleep paralysis) that are more common in NT2 than in idiopathic hypersomnia, may be used with the IHSS in the differential diagnosis [1, 33]. Although sleep inertia may be determined by responses to IHSS items 3–5 [9], the PVT and clinical interviews should also be conducted. For evaluation of habitual 24-hour sleep duration and EDS, and for distinguishing long sleepers from individuals with hypersomnia or idiopathic hypersomnia, self-report and objective measures should be used in a controlled manner, assessing EDS and sleep duration on both weekends and weekdays [75].

Another consideration is the requirement for ≤ 1 SOREMP, which may be an issue for patients who have idiopathic hypersomnia with MSLT latency > 8 minutes and sleep duration ≥ 11 hours on 24-hour PSG [1]. Individuals with ≥ 2 SOREMPs and MSLT latency ≤ 8 minutes meet the ICSD-3-TR criteria for narcolepsy diagnosis [2]. However, an idiopathic hypersomnia diagnosis should also be considered in patients who have ≥ 2 SOREMPs with MSLT latency > 8 minutes and total sleep time ≥ 11 hours. [1, 2] For individuals who have met the ICSD-3-TR criteria for an idiopathic hypersomnia diagnosis, and especially if HLA DQB1 0602 positivity is present, an assessment of cerebrospinal fluid (CSF) orexin levels may still be warranted, to further rule out a diagnosis of NT1.

Finally, depression may be an overlapping comorbidity in individuals with idiopathic hypersomnia [15, 17, 76]. A history of mixed anxiety-depression disorders is reported to be more common in individuals with idiopathic hypersomnia than in individuals with NT1 [11]. In the ARISE study, 12% of participants reported a comorbidity of major depression, while 67% of participants had scores ≥ 10 (reflecting moderate or severe depressive symptoms) on the Patient Health Questionnaire (PHQ-9) [15]. In a Swedish observational study, 28% of individuals with idiopathic hypersomnia had PHQ-9 scores ≥ 10 , and severity of depressive symptoms was a significant determinant of HR-QoL [17]. Evidence indicates a link between sleep inertia and depression [77], with strong correlations between IHSS scores, sleep inertia, depression, and anxiety observed in individuals with idiopathic hypersomnia [11]. Although a higher proportion of individuals with idiopathic hypersomnia who have LST (50%) than those who have normal nighttime sleep duration (36%) are reported to have psychiatric comorbidities [7], no significant differences in the rates or severity of depression and anxiety between LST and non-LST subgroups were observed [7, 11]. It should be noted that use of REM-suppressing antidepressant medications can reduce SOREMPs [78], which may result in a false negative for narcolepsy diagnosis or a false positive for idiopathic hypersomnia diagnosis on the MSLT. Therefore, if an individual's mood disorder is not severe enough to incur risk from temporary

treatment cessation, tapering antidepressant use to a full discontinuation for 2 weeks prior to MSLT, 6 weeks in the case of fluoxetine, is recommended to ensure an accurate diagnosis [78, 79].

In summary, the current diagnostic paradigm for idiopathic hypersomnia must be strengthened, as it may be allowing many cases of idiopathic hypersomnia to go undiagnosed. As EDS is not the only 24-hour symptom of idiopathic hypersomnia, inclusion of LST and sleep inertia assessments would bolster the diagnostic criteria and help avoid misdiagnoses. Additionally, use of the IHSS in real-world clinical settings should be assessed to further validate cutoff scores and MCID change scores in the context of successful treatment, comorbidities, and less severe forms of idiopathic hypersomnia [9].

Pathophysiology of idiopathic hypersomnia

The pathophysiology of idiopathic hypersomnia is unclear, as neurological biomarkers associated with sleep or wakefulness have not been identified in individuals with the disorder [1]. By definition, if measured, hypocretin 1 levels are normal in the cerebrospinal fluid (CSF) of individuals with idiopathic hypersomnia [80]. No significant differences in CSF histamine or *tele*-methylhistamine levels have been observed between people with idiopathic hypersomnia and people with other hypersomnia disorders [81, 82]. A study demonstrated that the CSF of individuals with idiopathic hypersomnia and with other hypersomnolence disorders showed enhanced γ -aminobutyric acid (GABA) receptor activity in the presence of exogenous GABA, indicating the possible presence of a substance that modulates the GABA receptor [83]. However, a subsequent study did not replicate these findings [84]. A prolonged circadian period or longer night phase with increased alertness during evening hours is seen in idiopathic hypersomnia [85, 86]. However, findings from a recent case-controlled study demonstrated no differences in the amplitude or phase of core body temperature circadian rhythm between individuals with idiopathic hypersomnia, healthy controls, and individuals with unspecified hypersomnia disorders [87]. In addition, core body temperature circadian rhythm did not appear to be associated with prolonged sleep duration or EDS [87]. There is a family history of hypersomnolence disorders in 34%–37% of individuals with idiopathic hypersomnia [1, 4, 29]. Whole-exome sequencing analysis of DNA identified 7 single nucleotide variants (SNVs) across 6 genes associated with sleep, wakefulness, and circadian rhythm in an LST subgroup of patients with established idiopathic hypersomnia [88]. An SNV of the *PER3* gene was confirmed in a separate group of patients with idiopathic hypersomnia with

LST [88]. During resting wakefulness in patients with idiopathic hypersomnia, altered regional cerebral blood flow characteristic of non-REM sleep has been observed, suggesting a persistent non-REM sleeplike state [89]. Increased metabolic activity in cortical brain regions constituting the salience network has also been observed [90].

Effects of treatments on 24-hour symptoms

Timely and effective treatment of daytime and nighttime symptoms is necessary to improve the health and well-being of individuals with idiopathic hypersomnia. The effects of various treatments on the 24-hour symptoms of idiopathic hypersomnia and their safety information are outlined in Table S1. Most of these therapeutics, primarily modafinil, clarithromycin, pitolisant, and stimulants (eg, methylphenidate, amphetamines), have been used off-label [91]. While some of these treatments have demonstrated some efficacy in treating EDS, their impact on other daytime or nighttime symptoms has not been assessed or reported. To date, LXB is the only medication approved by the US Food and Drug Administration (FDA) for the treatment of adults with idiopathic hypersomnia, and it has demonstrated efficacy for multiple symptoms beyond EDS [92]. An understanding of the strength of evidence for the efficacy of agents used to treat idiopathic hypersomnia symptoms can be beneficial in informing treatment decisions.

LXB was approved by the FDA for adults with idiopathic hypersomnia [92, 93] based on findings from a phase 3 randomized withdrawal study [41, 94]. In this trial, patients with idiopathic hypersomnia took LXB during an open-label titration/optimization period (OLT; 10–14 weeks) and a stable-dose period (SDP; 2 weeks) and then were randomized to LXB or placebo during a double-blind randomized withdrawal period (DBRWP; 2 weeks) [41]. LXB treatment during the OLT and SDP was associated with improvement in several daytime and nighttime idiopathic hypersomnia symptoms and HR-QoL (Table S1) [41, 94]. Improvements were also observed in ESS, IHSS, and VAS-SI scores [41]. During the DBRWP, ESS, IHSS, and VAS-SI scores increased (worsened) for patients who switched to placebo but were maintained for patients who continued LXB (Table S1) [41]. The proportions of patients who reported moderate or severe symptoms (ie, sleep inertia, long or frequent naps, long nighttime sleep duration, and cognitive dysfunction/brain fog) decreased with continued LXB treatment during the DBRWP [94]. The proportions of patients who reported moderate, significant, or very significant difficulty performing daily activities or driving also decreased with continued LXB use [94]. Although these findings strongly support use of LXB in individuals with idiopathic hypersomnia,

objective measures of sleep parameters (Maintenance of Wakefulness Test [MWT], PSG) were not incorporated. It is important to include these objective measures in future studies. Lastly, a high level of satisfaction with LXB treatment (mean 9-Item Treatment Satisfaction Questionnaire for Medication score, 69.8) was reported in an ongoing real-world study of individuals with idiopathic hypersomnia, with most (72.7%) reporting being satisfied or very satisfied with symptom relief [95].

Before LXB was approved, the AASM guidelines assigned a strong recommendation (one to be followed by clinicians in most circumstances) for use of modafinil in adults with idiopathic hypersomnia because its benefits outweighed its risks in this patient population [96]. Three randomized trials have evaluated the effect of modafinil in patients with idiopathic hypersomnia [97–99]. Across studies, modafinil has been shown to reduce EDS, number of daily naps, and overall symptom severity in individuals with idiopathic hypersomnia and to improve their driving ability [4, 96–102]. A randomized noninferiority trial of modafinil versus amphetamine-dextroamphetamine demonstrated similar improvements in EDS with both agents in participants with idiopathic hypersomnia or NT2 [102]. However, a higher proportion of participants treated with modafinil (31.8%) than with amphetamine-dextroamphetamine (9.1%) discontinued the study due to adverse events [102]. In a study of Japanese participants with idiopathic hypersomnia, 3 weeks of treatment with modafinil resulted in increases in sleep latency, based on the MWT, and decreases in ESS scores and number of daytime naps (Table S1) [97]. Additional randomized studies of modafinil had small populations and were not adequately powered [98, 99]. A proof-of-concept study of German participants with idiopathic hypersomnia demonstrated reductions in ESS scores after 3 weeks of treatment with modafinil; however, there was no improvement in MWT sleep latency [98]. In a French randomized study that evaluated automobile driving ability in patients with idiopathic hypersomnia or narcolepsy and healthy controls, treatment with modafinil was associated with increased MWT sleep latency as well as fewer inappropriate crossings into adjacent lanes and a trend toward less weaving [99]. In a retrospective single-center study of patients with idiopathic hypersomnia, modafinil reduced symptom severity [4]. Owing to the retrospective nature of the study, however, data on the impact of modafinil on other symptoms of idiopathic hypersomnia were not available [4]. Another retrospective study reported that modafinil resulted in a 4-point decrease in ESS scores in 62% of patients with idiopathic hypersomnia [29].

Evidence suggests that clarithromycin may improve EDS, cognition, fatigue, and the HR-QoL domains of physical function and daily activity [101, 103]. As demonstrated

in a study of patients with GABA-related hypersomnias, including idiopathic hypersomnia, clarithromycin improved scores on the ESS, the Functional Outcomes of Sleep Questionnaire–10 (FOSQ-10), and the Energy subscale of the 36-Item Short Form Health Survey (Table S1) [103]. While improvement in PVT reaction time (primary endpoint) was not seen with clarithromycin in the overall study population [103], a reduction in reaction time was reported in individuals with idiopathic hypersomnia [101]. The occurrence of dysgeusia or dysosmia in two-thirds of patients treated with clarithromycin may have contributed to unblinding and placebo effects [103]. The AASM guidelines assigned a conditional recommendation (one based on clinician's knowledge and experience, and consideration of patient's values and preferences) for use of clarithromycin in patients with idiopathic hypersomnia [96].

Methylphenidate, pitolisant, flumazenil, and amphetamines have been shown to improve EDS (decreases in ESS scores), although evidence for these agents was derived from retrospective analyses rather than randomized controlled trials (Table S1) [96, 100, 101, 104, 105]. In a retrospective study, 63% of patients with idiopathic hypersomnia treated with methylphenidate alone had a complete response, and 33% had a partial response (based on subjective descriptions by patients or clinicians, from clinical notes) [4]. A retrospective study that evaluated pitolisant in patients with idiopathic hypersomnia showed improved ESS scores in those with LST and without LST [104]. Flumazenil was evaluated in a retrospective study of patients diagnosed with various hypersomnolence disorders, including idiopathic hypersomnia [105]. Improved ESS scores were reported in 62.8% of patients; response rates were higher in female versus male patients (73% vs 48%) and in patients with versus without sleep inertia (72% vs 42%) [105]. However, outcomes specific to patients with idiopathic hypersomnia were not reported. Another retrospective study, which included patients with narcolepsy or idiopathic hypersomnia, reported an ESS score change of 3.1 with methylphenidate, which is similar to that observed with amphetamine (3.3) but lower than that observed with modafinil (6.0) [100]. Outcomes specific to patients with idiopathic hypersomnia were not reported. There is little published evidence on the impact of methylphenidate, pitolisant, flumazenil, and amphetamines on idiopathic hypersomnia symptoms beyond EDS.

Therapy that combines agents having different mechanisms of action may improve symptoms in individuals with central hypersomnolence, but the benefit is somewhat limited, and most evidence is from retrospective studies [4, 29, 100, 106]. In a study of patients with narcolepsy or idiopathic hypersomnia, approximately 18% of those with a partial response to monotherapy experienced a complete response

after switching to combination therapy; EDS improvement occurred in 38% of patients after switching to combination therapy [100]. In another study, improvements in daytime sleepiness and symptoms of depression were observed with both monotherapy and combination therapy in patients with idiopathic hypersomnia [106]. Reductions in total sleep time and depression symptom severity (from moderate to mild) were observed in patients who received combination therapy [106]. However, the study did not specify the monotherapy or combination therapies that were used.

In a recently published single-center, placebo-controlled, randomized, phase 3 study of participants with idiopathic hypersomnia, treatment with sodium oxybate (SXB) for 8 weeks significantly reduced EDS on the ESS and disease severity based on the IHSS score; improvements in wakefulness on the MWT and in HR-QoL were also observed [107]. This study was also the first to assess SXB treatment using PSG and MWT in individuals with idiopathic hypersomnia. However, the high sodium content (1100–1640 mg) in SXB at the recommended doses [108] should be considered, especially in patients with hypertension or other cardiovascular comorbidities [109]. Additionally, SXB is not approved by the US FDA for the treatment of idiopathic hypersomnia.

Given that most treatments for idiopathic hypersomnia have not been shown to improve many symptoms beyond EDS, it is not surprising that real-world studies of people with idiopathic hypersomnia demonstrated dissatisfaction with medications that were used off-label before LXB was approved. In the HFR study, despite treatment with off-label medications, >60% of individuals with idiopathic hypersomnia reported they continued to experience EDS, difficulty awakening, and difficulty functioning with a normal level of alertness [8]. In the ARISE study, individuals with idiopathic hypersomnia reported low treatment satisfaction, attributed primarily to lack of effectiveness of off-label medications [27].

Additional therapies are under investigation for the treatment of idiopathic hypersomnia. An ongoing placebo-controlled, randomized withdrawal study (NCT06525077) is evaluating the efficacy and safety of FT218 in adult participants with idiopathic hypersomnia. Several selective orexin 2 receptor agonists are being investigated for idiopathic hypersomnia, including ORX750, ALKS 2680, and TAK-360. A phase 2a study (NCT06752668) was initiated to investigate safety, pharmacokinetics, and efficacy of ORX750 in participants with NT1, NT2, and idiopathic hypersomnia [107]. An ongoing phase 2, placebo-controlled, dose-range-finding study (NCT06843590) is evaluating the safety and efficacy of once-daily ALKS 2680 in adult participants with idiopathic hypersomnia [110]. Another phase 2, dose-finding, placebo-controlled study (NCT06812078) is evaluating the safety, tolerability, and efficacy of TAK-360 in participants with idiopathic

hypersomnia. Solriamfetol, a dopamine and norepinephrine reuptake inhibitor, with doses ranging from 75–300 mg per day, is being evaluated (NCT06590662) for safety and efficacy in adults with idiopathic hypersomnia in a randomized, double-blind, placebo-controlled, phase 2 study [111].

Conclusions and future directions

Idiopathic hypersomnia symptoms are heterogeneous, occur continually day and night, and have a profound impact on health and well-being. This review adds to the current body of scientific literature on idiopathic hypersomnia by highlighting the 24-hour nature of the disorder, challenges associated with current diagnostic methods, and expert opinion perspectives on improving diagnosis to better inform treatment selection and improve patient outcomes. Although more research is greatly needed to identify reliable disease biomarkers to better understand the pathophysiology of this disorder, having a comprehensive understanding of idiopathic hypersomnia is paramount for accurate diagnosis and decreasing the likelihood of missed diagnosis. Current diagnostic criteria and disease assessment tools are limited in their ability to capture all the symptoms of idiopathic hypersomnia including LST. As well, better diagnostic criteria for narcolepsy may reduce the likelihood of a false-positive diagnosis of idiopathic hypersomnia and provide a more accurate clinical profile of idiopathic hypersomnia that is clearly differentiated from NT2. Modifications in diagnostic and disease management guidelines should be carefully considered and implemented to ensure more accurate diagnosis and inform optimal treatment strategies. A protocol that includes extended PSG (eg, a 24-hour sleep study) should be established and shared between institutions and used in home settings with actigraphy devices. Establishing new or modified billing strategies for such studies is necessary to enable implementation of extended PSG studies across clinical sleep laboratories. Use of the IHSS with the MSLT, ESS, and other validated symptom measures (SIQ, FSS, PHQ-9, PVT, FOSQ-10) and patient interviews to fully capture the 24-hour symptoms of idiopathic hypersomnia, both in clinical practice and as an endpoint in clinical and real-world studies, will advance the understanding of idiopathic hypersomnia and the impact of treatments. IHSS cutoff scores that can be used to distinguish idiopathic hypersomnia from NT2 and account for comorbidities should be identified; MCIDs that can be used to determine whether treatment is successful should be established.

Expert opinions

- Signs and symptoms of idiopathic hypersomnia occur throughout the day and night
- Fatigue, anxiety, and depression are frequently observed in individuals with idiopathic hypersomnia and should be assessed clinically with validated questionnaires
- Future studies should include the IHSS as an endpoint, which will refine our understanding of idiopathic hypersomnia symptoms and treatment response
- Better use of current diagnostic tools for idiopathic hypersomnia may help avoid inappropriate or suboptimal treatment, lack of treatment stemming from delays in diagnosis, or associated declines in QoL over the long term
- The IHSS should be evaluated in real-world clinical settings to establish cutoff scores that are applicable in clinical practice and account for successful treatment, comorbidities, and less severe forms of idiopathic hypersomnia

Abbreviations

Δ	Change
AASM	American Academy of Sleep Medicine
ARISE	Real-World Idiopathic Hypersomnia Outcomes Study
CSF	Cerebrospinal fluid
DBRWP	Double-blind randomized withdrawal period
EDS	Excessive daytime sleepiness
ESS	Epworth Sleepiness Scale
EQ-5D VAS	EQ-5D visual analog scale
FDA	Food and Drug Administration
FOSQ-10	10-Item Functional Outcomes of Sleep Questionnaire
FSS	Fatigue Severity Scale
GABA	γ-Aminobutyric acid
GABA _A	γ-Aminobutyric acid subtype A
HFR	Hypersomnia Foundation Registry
HR-QoL	Health-related quality of life
HSI	Hypersomnia Severity Index
IHSS	Idiopathic Hypersomnia Severity Scale
ICSD-3-TR	<i>International Classification of Sleep Disorders – Third Edition, Text Revision</i>
ILC	Illegal lane crossing
ITT	Intention to treat
LS	Least squares
LST	Long sleep time
LXB	Low-sodium oxybate
MCID	Minimum clinically important difference
MSLT	Multiple Sleep Latency Test
MWT	Maintenance of Wakefulness Test
NS	Not significant
NSS-2	Narcolepsy Severity Scale–2
NT1	Narcolepsy type 1
NT2	Narcolepsy type 2
OLT	Open-label titration/optimization period

PHQ-9	9-Item Patient Health Questionnaire
PSG	Polysomnography
PVT	Psychomotor Vigilance Task
REM	Rapid eye movement
SDLP	Standard deviation of lateral position
SD	Standard deviation
SDP	Stable-dose period
SF-36	36-Item Short Form Health Survey
SIQ	Sleep Inertia Questionnaire
SNV	Single nucleotide variant
SOREMP	Sleep-onset rapid eye movement period
SXB	Sodium oxybate
US	United States
VAS-SI	Visual Analog Scale for Sleep Inertia
WPAI:SHP	Work Productivity and Activity Impairment Questionnaire: Specific Health Problem

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s44470-025-00007-1>.

Acknowledgements Under the direction of the authors, Peloton Advantage, LLC (an OPEN Health company) employees Kalpana Vijayan, PhD, and Emily C. Bruggeman, PhD, provided medical writing support and an editor provided editorial support, which were funded by Jazz Pharmaceuticals.

Authors' contributions All authors were involved in the drafting, reviewing, and revision of this manuscript.

Funding This review was sponsored by Jazz Pharmaceuticals.

Data availability Not applicable.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication All authors have seen and approved the manuscript and consent to its publication.

Clinical trial registration This review does not report new data from a clinical trial.

Trial name, URL, ID number All N/A.

Prior presentation None.

Competing interests N **Foldvary-Schaefer** has received research grants from Avadel, Jazz Pharmaceuticals, Harmony, Suven, Takeda, and Vanda; served on a publication committee and advisory committees for Jazz Pharmaceuticals; and received royalties from UpToDate and Oxford University Press. **K Maski** has received personal compensation for serving on advisory boards for Alkermes, Avadel, Eisai, Harmony Biosciences, Jazz Pharmaceuticals, and Takeda. She receives research support from Harmony Biosciences, Jazz Pharmaceuticals, and Takeda. She receives research support from National Institutes of Health, National Institute of Neurological Disorders and Stroke under award number R61NS130215-01A1. **LD Schneider** has received

personal compensation for serving on advisory boards and speakers bureaus for Avadel, Eisai, and Jazz Pharmaceuticals. **Y Dauvilliers** is a consultant for and has participated in advisory boards for Avadel, Bioprojet, Centessa, Harmony Biosciences, Idorsia, Jazz Pharmaceuticals, and Takeda.

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